

Student Name: Jinpeng Zhu  
Student ID: 1907843  
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## Underground Mining and Bulk Materials Handling (MIN E – 630) Assignment #4

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### 1 Question 1

RQD: The method for measuring rock quality refer the (Deere 1988)

#### 1.1 Shape factor (HR) and the RQD

Refer (Norwegian Geotechnical Institute 2025)

For hanging wall and foot wall:

$$HR = \frac{80 \times 45}{2(45 + 80)} = 14.4$$

For crown:

$$HR = \frac{30 \times 45}{2(45 + 30)} = 9$$

For the end wall

$$HR = \frac{80 \times 30}{2(30 + 80)} = 10.9$$

Based on the RQD, Jn, Jr, and Ja, we can get the  $Q'$

$$Q' = \frac{RQD}{J_n} \times \frac{J_r}{J_a} = 60.5$$

The stability number (N) is related to  $Q'$ , A, B, and C, as shown in the Equation 1

$$N = Q' \times A \times B \times C \quad (1)$$

Where: - A is the measure of the ratio of intact rock strength to induced stress. As the maximum compressive stress acting parallel to a free stope face approaches the uniaxial compressive strength of the rock, factor A degrades to reflect the related instability due to rock yield. - B is the measure of the measure of the relative orientation of dominant jointing with respect to a free stope face. Joints forming shallow angle (10-30 degrees) with the free face are likely to become unstable, where joints perpendicular to the free face have little influence on stability. - C is the measure of the influence of gravity on the stability of the face being considered. The back of the stop or structural weaknesses of a stop oriented unfavorably with respect to gravity sliding have a maximum impact on stability.

The data charts of A, B and C are from (Potvin 1988), also refer to the (Mawdesley, Trueman, and Whiten 2001)

## 1.2 A Rock stress factor

To determine the stress factor for a surface the induced stresses.

The vertical in-situ stress is :

$$\sigma_V = \eta Z = 2.3 \times 1000 \times 9.81 \times 1500 = 33.84 MPa$$

$$\sigma_{H1} = K \times \sigma_V = 33.84 \times 1.6 = 54.15 MPa$$

$$\text{Thus } \sigma_{H2} = \frac{\sigma_{H1}}{1.1} = 49.23 MPa$$

### 1.2.1 Take the Crown as an example

Induced stress at top of the vertical plane: Surface height = 80; surface span = 30. height to span ratio: 2.67, K = 1.45.

For a height to span ratio of 2.67, and a K value of 1.45.  $\sigma_1/\sigma_V$  is estimated at 2.7. From this relationship it can be calculated that:

$$\sigma_1 = 2.7 \times 33.84 = 91.37 MPa$$

Therefore, the UCS of the rock to the induced stresses can be calculated:

$$\frac{\sigma_c}{\sigma_1} = 100/91.37 = 1.096 < 2$$

Per the relationship figure of rock stress factor, The value of factor A is 0.1.

Similarly, the A factor for the hanging wall, foot wall and end wall can be calculated as 1, 0.15, 0.2 respectively, as shown in the Figure 1.

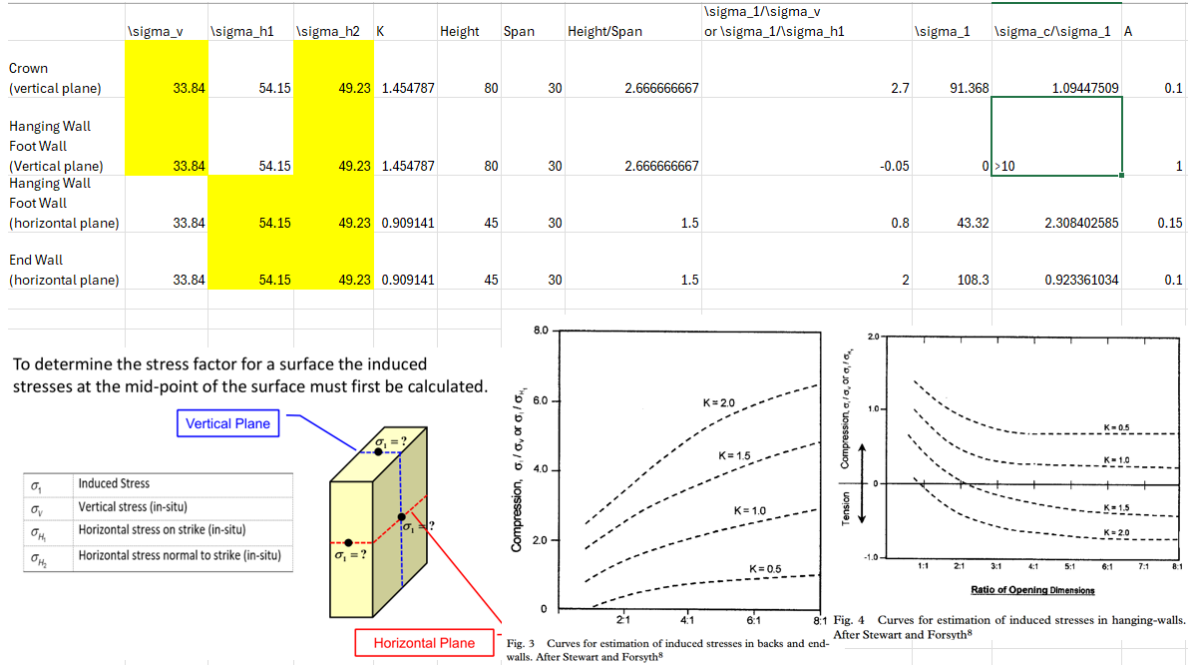


Figure 1: A: Rock stress factor

Note: the A factor for the foot wall and hanging wall has two possible values (two planes), the lower value is used to be conservative, so we use 0.15 to calculate A.

### 1.3 B and C: Joint orientation factor and Design surface orientation factor

The dips of the open stope is 65 degree, the dip of the joint set is 20 degree, so the angle between the joint set and the free face is 45 degree. we can get the B factor from the right side of the Figure 2.

As shown in the Figure 2, the B and C factors for the hanging wall, foot wall, crown and end wall can be determined as 0.3, 0.4, 0.4, 1 respectively.

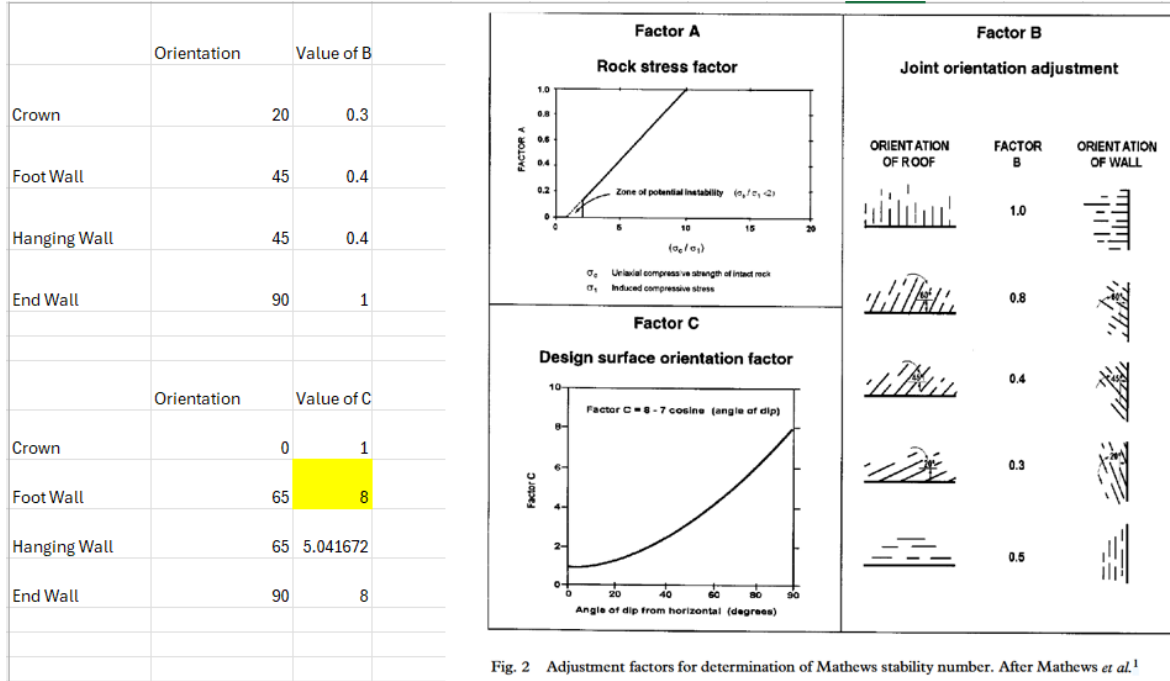


Fig. 2 Adjustment factors for determination of Mathews stability number. After Mathews *et al.*<sup>1</sup>

Figure 2: B and C

After get the A, B and C factors, we can calculate the stability number N, and marking the stability of the stope as shown in the Figure 3.

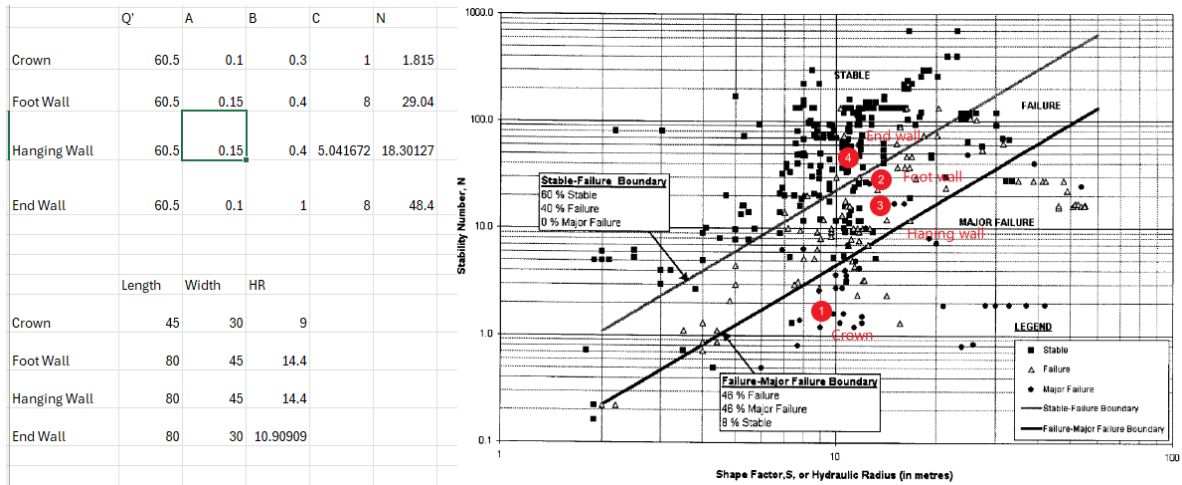


Figure 3: Stability of the slope

The end wall is most stable, the hanging wall and foot wall are in failure zone, and the crown is in the major failure zone.

## 2 Caveability

The solutions to this question referred from (Laubscher 1990; Hustrulid and Bullock 2001)

UCS: Due to only know the single UCS, assume the rock mass is homogenous, the rating is 9 from the table 1 in (Laubscher 1990).

Joint condition: the moist and moist pressure unknown, ignore it.

Through the MRMR method, the IRMR is 59.92, the MRMR is 41.94, and the minimum threshold area is  $5400 m^2$ . The computation process please see the marking on the problem statement, and the original source basis please see the attached pdf file.

## 3 References

- Deere, D. 1988. "The Rock Quality Designation (RQD) Index in Practice." In *Rock Classification Systems for Engineering Purposes*, 91–101. ASTM International.
- Hustrulid, W. A., and Richard L. Bullock. 2001. *Underground Mining Methods: Engineering Fundamentals and International Case Studies*. SME.
- Laubscher, D. H. 1990. "A Geomechanics Classification System for the Rating of Rock Mass in Mine Design." *Journal of the Southern African Institute of Mining and Metallurgy* 90 (10): 257–73. [https://doi.org/10.10520/AJA0038223X\\_1954](https://doi.org/10.10520/AJA0038223X_1954).
- Mawdesley, C., R. Trueman, and W. J. Whiten. 2001. "Extending the Mathews Stability Graph for Open-Stope Design." *Mining Technology* 110 (1): 27–39. <https://doi.org/10.1179/mnt.2001.110.1.27>.
- Norwegian Geotechnical Institute. 2025. *Using the Q-system: Rock Mass Classification and Support Design*. Manual. Oslo, Norway: NGI.
- Potvin, Yves. 1988. "Empirical Open Stope Design in Canada." PhD thesis, University of British Columbia. <https://doi.org/10.14288/1.0081130>.

## MINE E 630: Assignment #4

**Due Date:** March 12, 2026 (until 11: 59 pm)

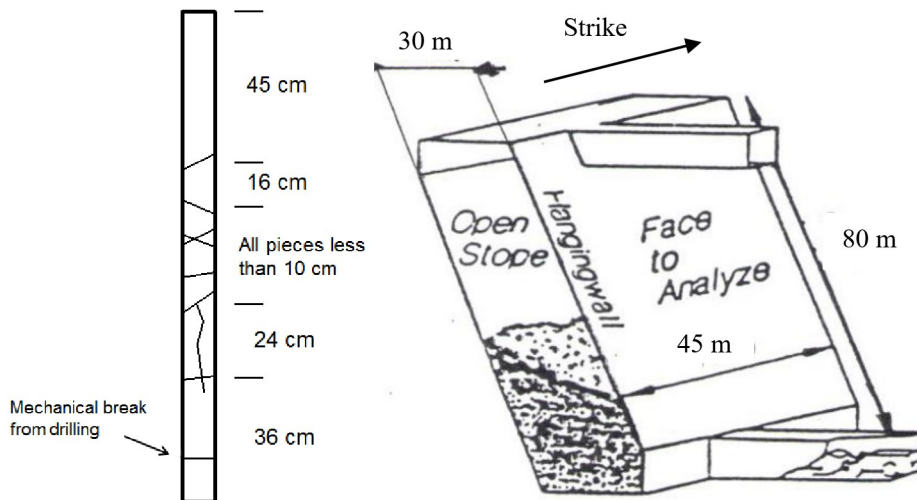
Method of Submission: Please write down your words and submit a combined pdf file to the Canvas assignment before the specified due time.

### 1. Open Stope design (Please submit your question sheets together with your answer sheets)

An open stope is planned at a depth of 1500 m in an orebody that is 30 m wide and dips at 65°. Because of operational constraints, the preferred stope length is 45 m, and the stope height is 80 m.

The average unconfined compressive strength of the intact rock is 100 MPa. There is one joint set dipping 20° (closely spaced, average spacing is 10 cm) with the same strike direction as the orebody strike.

A typical drill core (the total length of the core is 200 cm) and the stope geometry have been shown below:



Rock mass quality data has been presented in the following table (Question marks in the table are to be answered):

| Item             | Description                     | Value | RQD = (45+16+24+36)/200 * 100% = 60.5% |
|------------------|---------------------------------|-------|----------------------------------------|
| Rock quality     | ? Fair                          | ?     |                                        |
| Joint sets       | One joint set                   | ?     | 2                                      |
| Joint roughness  | Smooth undulating               | ?     | 2                                      |
| Joint alteration | Unaltered with surface staining | ?     | 1                                      |

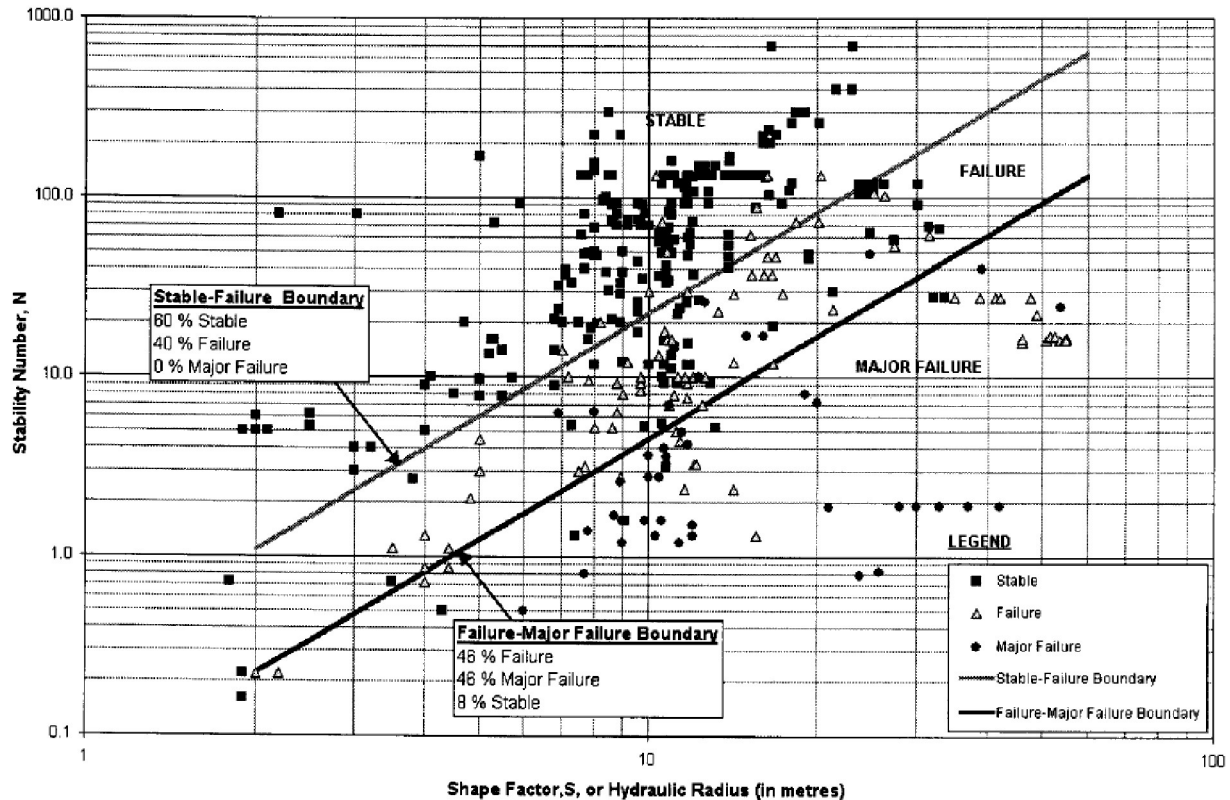
$$\sigma_{H1} = 1.1\sigma_{H2}$$

Along to the strike direction

$$K = \frac{\sigma_{H1}}{\sigma_V} = 1.6 \quad \text{perpendicular to the } \sigma_{H1}$$

The average specific gravity of overburden above the slope is 2.3.

Please estimate the stability of this open slope from the attached Matthews stability graph. Please mark the point and name of each slope surface on this following Matthews Chart with a colour pen.

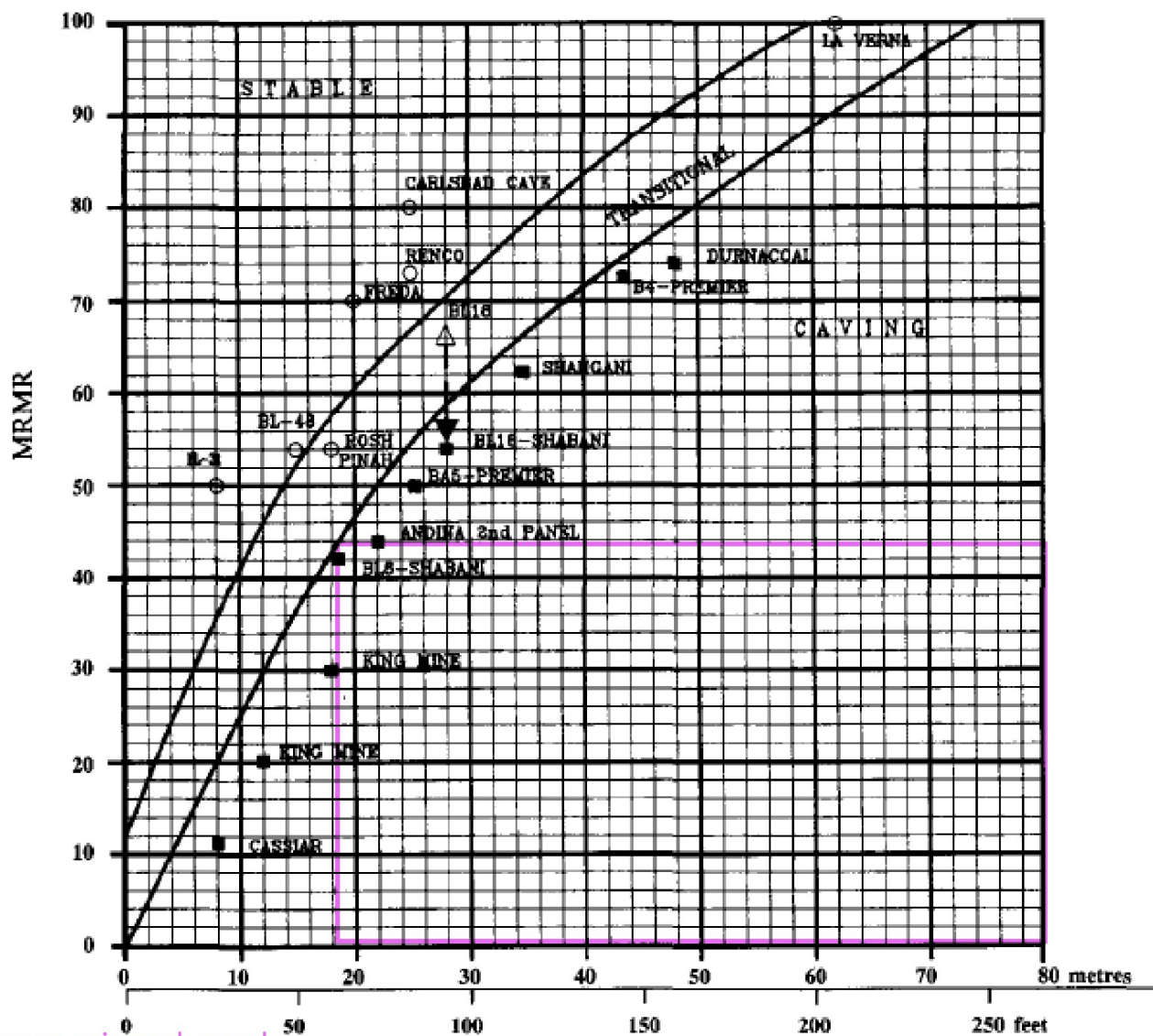


## 2. Caveability

The Laubscher Stability Chart relates the rock quality and stress conditions for a given deposit, characterized by the Modified Rock Mass Rating (MRMR), to the hydraulic radius of the opening. If **block caving** is considered as the mining method, please estimate the minimum undercutting area (length: width = 1.5) to be required. Please get numbers for all question marks, illustrate your calculations and show your procedure in the attached Laubscher Stability Chart.

| Parameter       | Description                           | Rating                                             |
|-----------------|---------------------------------------|----------------------------------------------------|
| UCS             | 90 MPa                                | ? 9                                                |
| RQD             | 80%                                   | ? 12                                               |
| FF/m            | 3 joint sets, average spacing of 0.5m | ? $25 \times 0.7 = 17.5$                           |
| Joint condition |                                       | ? $= 40 \times 0.85 \times 0.9 \times 0.7 = 21.42$ |
| Large scale     | Dry, curved                           | 85%                                                |
| Small scale     | Dry, smooth stepped                   | 90%                                                |

|                         |                              |                           |
|-------------------------|------------------------------|---------------------------|
| Joint wall alteration   | None                         |                           |
| Joint filling           | Soft sheared coarse material | 70%                       |
| <b>IRMR</b>             |                              | ? $9+12+17.5+21.42=59.92$ |
| Adjustments             |                              |                           |
| Weathering              | None                         | 100%                      |
| Joint orientation       | 3 joints, 3 inclined         | 70%                       |
| Mining induced stresses |                              |                           |
| Blast effects           | None                         | 100%                      |
| <b>MRMR</b>             |                              | ? $59.92*1*0.7*1=41.94$   |



To cave mineral needs  
 $HR > 18$  .

**HYDRAULIC RADIUS = Area/perimeter**

So, set width = x, length =  $1.5x$   $1.5x/5 > 18 \rightarrow x > 60$ , set  
 $x = 60$  thus the length = 90, so the minimum threshold area is  
 $5400m^2$